Venkata Lakshmi Parimala Pasupuleti

Software Engineer · AI for Housing & Healthcare Operations · Python · Java · Microservices · AWS · Event-Driven · Production AI

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SUMMARY

Software engineer with **7+ years of experience** building production systems in the two industries EliseAI serves - **housing and healthcare**. At **Entrata** (7 years, affordable housing SaaS, 100K+ users): owned the full resident lifecycle across HUD, Tax Credit, and HOME programs - eligibility, certifications, recertifications, rent calculations, gross rent changes, HAP processing, resident management, and the annual HUD income limits import (180K records, 7 consecutive cycles). Owned the resident document migration to AWS cloud. Understand housing operations from the inside, not just the API layer. Currently at Intuit: **AI agents in production** with tool calling, structured outputs, HITL guardrails, and CI eval harnesses. Built **Leasel** - a production lease compliance AI platform: 8-agent LangGraph orchestration, document RAG (OCR, hybrid search, citation grounding), HITL guardrails, and a **CI eval harness** (93.6% precision, 1,240 test cases). Built **AI on FHIR** - a clinical AI system connecting natural-language queries to healthcare APIs (FHIR REST, LLMs, BERT, HIPAA-style access controls) - direct healthcare domain coverage. Strong in **Python, Java, FastAPI, microservices, event-driven systems, AWS**. Ship fast, own end-to-end, stay with it after it ships. M.S. Data Analytics Engineering, George Mason University (GPA 3.87).

CORE COMPETENCIES

Housing Operations · Affordable Housing · HUD · Tax Credit · HOME · HAP

Leasing · Resident Management · Certifications · Rent Calculations · Compliance

Healthcare AI · Clinical Workflows · FHIR · Patient Data · HIPAA-Style Access

AI Agents · Autonomous Workflows · Tool Calling · LangGraph · Production AI

Python (7+ yrs) · Java (7+ yrs) · FastAPI · Spring Boot · TypeScript · React

Microservices · Event-Driven · Distributed Systems · RabbitMQ · SQS

AWS (S3 · Lambda · SQS · IAM · CloudWatch · API Gateway) · Docker · Kubernetes

Scalability · Reliability · Observability · CI/CD · GitHub Actions

End-to-End Ownership · Ship Fast · Production Habits · Startup Mindset

RAG · Document Understanding · OCR · LLM Evals · Guardrails · HITL

PostgreSQL · SQL · ETL · Data Pipelines · Async Processing

Agile · Code Review · Technical Documentation · Cross-Functional Collaboration

EXPERIENCE

Software Engineer

Jun 2026 - Present

Intuit — Virginia, USA (Remote)

Financial services AI - LLM-powered agents and workflows in production

- **Shipped AI agents end-to-end** in production: LangChain + Anthropic Claude with multi-step tool calling, structured outputs, retrieval-grounded context, and HITL for sensitive decisions - same pattern as EliseAI's autonomous building and healthcare workflows.
- **Built and owns the LLM eval harness:** CI regression gating with golden datasets and LLM-as-judge scoring; structured logging per agent run; failure analysis → eval coverage - ensuring agents stay reliable in production between every release.
- Build ML data pipelines (scikit-learn, XGBoost); React/TypeScript front ends; deploy via Docker + GitHub Actions CI/CD to AWS with CloudWatch observability. Deploy frequently, iterate fast.

Senior Software Engineer

Oct 2020 - Jul 2024

Entrata — Affordable Housing & Multifamily SaaS

Regulated multi-tenant platform · 100K+ monthly active users · HUD, Tax Credit, HOME affordable housing programs

- **Owned core housing operations workflows end-to-end:** annual HUD income limit imports (180K records, 7 consecutive cycles, zero violations); async rent update processing across thousands of tenant records via RabbitMQ (idempotency, retry logic, dead-letter routing); eligibility, certifications, recertifications, and gross rent change workflows across HUD, LIHTC, and HOME programs.
- **Owned the resident document migration to AWS cloud** - moved the full resident document repository from legacy storage to S3; led Docker microservices redesign (50% scalability improvement); optimized 200+ SQL queries (30% performance gain); Python/Bash automation eliminating 80% of manual work; 300+ code reviews; mentored 5+ engineers across 120+ Agile sprints.

Junior Software Engineer

Sep 2018 - Oct 2020

Entrata India — Pune

Property management platform - affordable housing and market-rate multifamily modules

- Built housing management modules (Python, PHP, Java, PostgreSQL); delivered large-scale data integration workflows (1.6M HUD/Tax Credit records); created 56 state-specific housing certification documents; led AWS S3 migration from legacy storage.

PROJECTS

Leasel - Lease Compliance AI Platform: Agents · RAG · Guardrails · Evals

2026 | Python, LangGraph, FastAPI, ChromaDB, MCP, CI/CD | github.com/venkatalakshmiparimala/Lease_complaince_rag

- Production lease compliance AI platform - directly in EliseAI's housing domain: **8-agent LangGraph orchestration** for autonomous multi-step document workflows (triage, retrieval, reasoning, validation); **document RAG** (OCR ingestion, hybrid TF-IDF + semantic reranking, citation-

grounded answers); **HITL guardrails** routing regulated decisions to staff; MCP server integration; **CI eval harness** (LLM-as-judge, 93.6% precision, 1,240 test cases). Iterated from prototype to 3 production surfaces - owned the full arc.

AI on FHIR — Healthcare Workflow Automation & Clinical API Integration

2024 | Python, Flask, React, BERT, LLMs, FHIR REST APIs, Docker | github.com/venkatalakshmiparimala/AI_on_FHIR

- Built an AI system that automates healthcare workflows - translates natural-language clinical questions into FHIR REST API calls using LLMs, BERT, and embeddings, cutting lookup time ~60%. **Directly relevant to EliseAI's healthcare vertical**: HIPAA-style RBAC, OAuth-secured clinical data integration, Docker-containerized. Demonstrates end-to-end ownership in both housing and healthcare domains.

AgentProbe - Agent Eval & Observability Platform 2025 | Python, FastAPI, PostgreSQL, CI/CD | github.com/venkatalakshmiparimala/AgentProbe

- Open-source platform for **production AI reliability**: offline golden-dataset evals, LLM-as-judge scoring, CI regression gating, failure classification; cost-aware model routing across 6 providers; full per-run tracing and observability. Used as a release gate before every AI deployment - because shipping 100+ times per day requires knowing what breaks.

EDUCATION

M.S. Data Analytics Engineering — George Mason University, Virginia | GPA 3.87 May 2026

B.Tech Computer Science & Engineering — JNTU Anantapur 2017